Evan Frangipane

+1 (310) 310-9459 frangipane.evan@gmail.com Los Angeles, CA

Linkedin evanfrang evanfrangipane.com

Summary

Ph.D. in Theoretical Physics with expertise in data analytics, experimentation, and machine learning. Experienced in designing A/B tests, developing statistical models, and leveraging data insights to drive decision-making. Proficient in Python (Pandas, NumPy, SciPy, scikit-learn), SQL, and data visualization tools. Skilled in exploratory data analysis, hypothesis testing, and building scalable data pipelines. Passionate about using data science to optimize customer experience, marketplace dynamics, and operational efficiency.

Education

Ph.D. in Physics *University of California, Santa Cruz* **Santa Cruz, CA** *2018-2024*

B.A. in Physics *University of California, Berkeley* **Berkeley, CA** *2014-2018*

Certifications

Google Advanced Data Analytics Professional Certificate *2025*

Employment

Graduate Student Researcher, *University of California, Santa Cruz* *2018 - 2024*

- Designed and executed Python-based data analysis pipelines to extract insights from large, complex datasets.
- Conducted hypothesis testing, causal inference analysis, and statistical modeling to support data-driven decision-making.
- Developed automated data pipelines using Python and SQL for efficient data processing.
- Collaborated with cross-functional teams to refine research methodologies and interpret results.

Teaching Assistant, *University of California, Santa Cruz* *2018 - 2024*

- Taught Python-based data analysis, covering statistical techniques, regression modeling, and machine learning applications.
- Mentored students in experiment design, hypothesis testing, and data visualization techniques.

Undergraduate Researcher *Lawrence Berkeley National Laboratory* *2016 - 2018*

- Participated in detector development for the ATLAS collaboration
- Contributed to detector development white paper for the High Luminosity Large Hadron Collider

Technical Projects

Multi-Model Fraud Detection: LogReg, SVC, and Random Forest (Self-directed) *2025*

- Compared Logistic Regression, Linear SVC, and Random Forest models for credit card fraud detection on a highly imbalanced (280k transactions, $< 0.2\%$ fraud) dataset.
- Utilized Python, Scikit-learn, matplotlib for data preprocessing, model development, evaluation, and visualization
- Random Forest outperformed logistic regression and SVC with an 0.84 AUCPR and 0.82 f1-score.

Multiple Testing Problem and the Bonferroni Correction (Self-directed) *2025*

- Simulated the multiple testing problem using Python to analyze how repeated statistical tests inflate false positive rates.
- Implemented Monte Carlo simulations and applied the Bonferroni correction to mitigate inflated significance levels.
- Visualized results with Matplotlib, comparing analytical predictions to numerical data and identifying discrepancies due to discrete effects.

Statistical Analysis of Primordial Black Hole Hypotheses *2023*

- Performed large-scale microlensing simulations using Kubernetes, efficiently running parallelized jobs on a distributed computing cluster to save hundreds of hours
- Applied Anderson-Darling statistical tests to differentiate population distributions of Free-Floating Planets and Primordial Black Holes
- Conducted extensive Exploratory Data Analysis (EDA) using Python (Pandas, NumPy, Matplotlib)

Skills

- **Languages:** Python (NumPy, Pandas, SciPy, Matplotlib, scikit-learn), SQL, Mathematica, C++
- **Statistical Analysis:** Hypothesis Testing, Regression, Confidence Levels, Probability Theory, A/B Testing, Machine Learning
- **Soft Skills:** Data Storytelling, Report Writing, Stakeholder Communication
- **Tools:** LaTeX, Git, Docker, Kubernetes, Jupyter Notebooks